

VIVEK SINGH

CFD Engineer | Fluid Dynamics, Aerodynamics | MSc Automotive Engineering

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SUMMARY

CFD engineer with over two years' hands on simulation experience across OpenFOAM, Ansys Fluent, and STAR CCM+, and a solid grounding in fluid mechanics, aerodynamics, and heat transfer through professional practice and MSc level research. Comfortable working across mesh generation, turbulence modelling, boundary condition specification, and post processing, and experienced in validating simulation and surrogate model predictions against independent reference data the same rigour needed to confirm an engineering problem is precisely defined and objectively solvable. Strong written communicator, having presented technical findings to multidisciplinary engineering teams and at an international research conference. Self directed by nature, MSc research is independently scoped and executed with minimal supervision.

AREAS OF EXPERTISE

CFD Solvers: OpenFOAM, Ansys Fluent, STAR CCM+, ANSA, Hypermesh | Fluid Mechanics, Aerodynamics & Heat Transfer | Mesh Generation & Turbulence Modelling | Boundary Condition Specification & Post Processing | Verification & Validation Against Reference Data | Programming: Python, MATLAB | CAD Modelling: CATIA V5, AutoCAD | Technical Writing & Cross Disciplinary Communication

PROFESSIONAL EXPERIENCE

Tata Technologies | Pune, India

CFD Engineer July 2023 – September 2025

- Performed CFD simulations in both OpenFOAM and Ansys Fluent to evaluate aerodynamic and thermal performance of vehicle systems, applying mesh generation, turbulence modelling, and boundary condition specification across each solver.
- Assessed and optimised aerodynamic performance through numerical modelling of airflow and heat transfer in HVAC and vehicle exterior domains.
- Supported design optimisation and concept validation, ensuring aerodynamic efficiency aligned with development targets.
- Collaborated in multidisciplinary reviews with design and thermal engineers, translating simulation results into clear, actionable engineering conclusions.
- Improved CFD workflow efficiency by 10% through process automation.
- Delivered data-driven insights and clearly communicated performance results, risks, and improvement opportunities to senior engineers.

EDUCATION

MSc Automotive Engineering

Cranfield University, Cranfield, UK | September 2025 – September 2026

- Modules: Vehicle Design; Propulsion and Performance; Automotive Control and Simulation; Vehicle Electrification and Hybridization; Vehicle Dynamics; Powertrain Simulation and Performance; Automotive Aerodynamics; Vehicle Structures; Vehicle Materials and Manufacturing.

B. Tech in Mechanical Engineering (Specialisation in Automobile Engineering)

DIT University, Dehradun, India | July 2019 – December 2023 - CGPA 7.4/10

KEY PROJECTS

Unsteady Aerodynamic Data Modelling —MSc Cranfield University(ongoing) | May 2026 – Sep 2026

- Developing Gaussian Process and surrogate models to predict unsteady aerodynamic data, aiming to reduce dependency on expensive high fidelity CFD runs and wind tunnel cost during concept screening.

- Validating surrogate predictions against wind tunnel experimental databases (hot wire anemometry, pressure, force/moment balance data) alongside DDES/LES numerical references, building direct CFD to test correlation experience.
- Running sensitivity studies to identify the parameters most influential on force coefficient prediction, deepening flow understanding and guiding future simulation campaign focus areas

L7e Quadricycle – Structural Integrity Assessment | Cranfield University GDP | March 2026 – April 2026

- Conducted LS DYNA crash simulations (frontal, side pole, and quasi static roof crush) on an EU L7e quadricycle chassis, evaluating energy absorption, peak contact forces, and stress distribution to assess passenger compartment integrity
- Identified critical structural failure modes from crash test data and proposed design improvements, including front crash boxes, lateral bracing, and reinforced A pillar cross sections
- Designed and modelled the vehicle chassis in CAD, including material selection via a weighted SAW (Simple Additive Weighting) decision matrix to compare steel and aluminium alloys against stiffness, cost, and weight criteria for EU regulatory compliance
- Performed FEA setup and analysis using Altair HyperMesh/OptiStruct, building 1D and 2D models (8,000+ elements) to evaluate torsional and bending stiffness, informing structural lightweighting decisions
- Presented crashworthiness and structural risk findings to a multidisciplinary, 7 person team and academic advisors as part of the Group Design Project

Formula Student Race Car - Aerodynamic Package Design | DIT University Parakram Team | 2019 – 2022

- Led the aerodynamic development of the Parakram Formula Student car as Team Captain, overseeing concept generation, CFD evaluation, and physical build of front wing, rear wing, and underbody elements.
- Generated and evaluated multiple aerodynamic concepts across front and rear of the vehicle, assessing downforce, drag, and aerodynamic balance trade offs to optimise overall car performance within Formula Student regulations.
- Drove the car in national competition, gaining first hand understanding of aerodynamic balance, on car behaviour, and the interaction between aero development targets and real world driver feedback.

Fabrication and Analysis of Hybrid Composite Material

- Fabricated a hybrid composite material and performed mechanical tests to examine its properties under different conditions.
- Integrated the fabricated material as the outer body for an electric vehicle prototype, considering aerodynamic and structural implications.
- Performed thermal analysis of the manufactured composite and analysed property variation with temperature.
- Presented research findings at an international conference, communicating complex results effectively.

TRAINING

- EV Thermal Management System — I-GET-IT (2023)
- CATIA V5 Fundamentals — I-GET-IT
- Fundamentals of GD&T — I-GET-IT

VOLUNTEERING EXPERIENCE

Parakram Formula Student Team, DIT University | India

Sep 2019 – Mar 2022

Team Captain & Development Driver

- Directed all technical and operational aspects of the Formula Student programme, managing a multidisciplinary team of engineers across aerodynamics, chassis, powertrain, and electronics.
- Developed and championed the aerodynamic concept strategy for the vehicle, driving the programme from initial idea generation through CFD evaluation to physical build and competition validation.
- Competed as development driver at national level, providing qualitative aerodynamic and vehicle balance feedback to inform further iteration closing the loop between simulation, design, and real-world performance.
- Secured external technical and financial sponsorships through professional stakeholder engagement, demonstrating commercial awareness alongside engineering capability